

King County Housing EDA Documentation

Summary of the exploratory data analysis.

Data Cleaning

- Filled missing values in waterfront, view and sqft_basement. - Restored missing values (NaN) in yr_renovated after review and corrected renovation years (19910→1991). - Converted date to datetime and zipcode to categorical. - No duplicate records were found.

Outliers

House prices are right-skewed. Luxury homes were treated as valid observations, so price outliers were retained.

Hypothesis 1: Location

Supported. Zipcode shows large differences in median price (about \$235k to \$1.895M). Waterfront homes (~\$1.51M median) are much more expensive than non-waterfront homes (~\$450k). Better view ratings are generally associated with higher prices.

Hypothesis 2: Renovation

Supported. Renovated homes have a higher median price (~\$607,502) than non-renovated homes (~\$448,000).

Hypothesis 3: Grade

Supported. Median price increases strongly as grade increases.

Hypothesis 4: Seasonality

Weak / Not supported. Monthly median prices vary only modestly (about \$426k–\$477k).

Overall Conclusion

Supported: 1. Location affects house prices. 2. Renovation is associated with higher prices. 3. Higher grade is associated with higher prices. Weak evidence: 4. Seasonality has only a small effect compared with the other factors.